Maximum and minimum values

1 Consider the function $f(x) = 2x^3 - 12x^2 + 18x + 3$.

a Find the x-coordinates of the turning points.

b State the coordinates of the local maximum.

c State the coordinates of the local minimum.

d State the absolute maximum value on $[0, 7]$.

e State the absolute minimum value on $[0, 7]$.

2 Consider the function $y = 5x^3 + 7x^2 + 3x + 4$ with restricted domain of $[-1, 1]$.

a Find the y-intercept.

b Find the stationary points.

c Determine the nature of the stationary points.

d Find any points of inflection and classify them.

e Describe the behaviour of the function as $x \to \pm\infty$.

f Find the maximum and minimum value of the function for the restricted domain.

3 Consider the function $y = (x^2 - 1)^2 + 4$ with restricted domain of $[-2, 2]$.

a Find the y-intercept.

b Find the x-intercept(s).

c Determine whether the function is odd or even.

d Find the stationary points.

e Determine the nature of the stationary points.

f Find any points of inflection and classify them.

g Describe the behaviour of the function as $x \to \pm\infty$.

h Find the maximum and minimum value of the function for the restricted domain.

4 Consider the function $y = 3x^4 + 4x^3 - 12x^2 - 1$.

a Find the y-intercept.

b Find the stationary points.

c Determine the nature of the stationary points.

d Find any points of inflection and classify them.

e Describe the behaviour of the function as $x \to \pm\infty$.

f Find the maximum and minimum value of the function for the restricted domain $[-1, 1]$.

© ClipSAT: By Mr. Mohamed Abdallah +977597688647

Graphs of functions

5 Consider the function $y = -x^2 + 8x - 21$ with restricted domain of $[0, 6]$.

- **a** Find the y-intercept.
- **b** Find the stationary points.
- **c** Determine the nature of the stationary points.
- **d** Find the maximum and minimum value of the function for the restricted domain.
- **e** Sketch the function on a number plane.

6 Consider the function $y = x^4 - 2x^2 + 1$ with restricted domain of $[-2, 3]$.

- **a** Find the y-intercept.
- **b** Find the x-intercept(s).
- **c** Determine whether the function is odd or even.
- **d** Find the stationary points.
- **e** Determine the nature of the stationary points.
- **f** Find any points of inflection and classify them.
- **g** Find the maximum and minimum value of the function for the restricted domain.
- **h** Sketch the function on a number plane.

7 Consider the function $y = (2x - 1)^2(1 - x)$ with restricted domain of $[0, 2]$.

- **a** Find the y-intercept.
- **b** Find the x-intercept(s).
- **c** Find the stationary points.
- **d** Determine the nature of the stationary points.
- **e** Find any points of inflection and classify them.
- **f** Describe the behaviour of the function as $x \to \pm\infty$.
- **g** Find the maximum and minimum value of the function for the restricted domain.
- **h** Sketch the function on a number plane.

8 Consider the function $y = 9x^2 + 18x - 16$.

- **a** Find the y-intercept.
- **b** Find the x-intercept(s).
- **c** Find the stationary points.
- **d** Determine the nature of the stationary points.

© ClipSAT: By Mr. Mohamed Abdallah +977597688647

9 Consider the function $y = 8(x-1)^2(x+2)$.

a Find the stationary points.

b Determine the nature of the stationary points.

c Determine the point of inflection.

d Sketch the function on a number plane.

10 For each of the following functions:

i Find any stationary points and determine their nature.

ii Determine the points of inflection.

iii Sketch the graph showing stationary points and points of inflection.

iv Write the domain for which the function is decreasing.

a $f(x) = 8 - x^3$

b $f(x) = 8 - x - x^3$

c $f(x) = x^3 - 3x^2 - 9x + 11$

d $y = x^3 - 6x^2 - 3$

e $f(x) = \left(x^2 - 9\right)^2 + 3$

f $f(x) = x^3 - 5x^2 + 3x - 5$

11 For each of the following functions:

i Find any stationary points and determine their nature.

ii Determine the points of inflection.

iii Sketch the graph showing stationary points and points of inflection.

iv Write the domain for which $\dfrac{dy}{dx}$ is positive.

a $f(x) = x^4(3x - 10)$

b $f(x) = x^2(x + 3)$

12 For each of the following functions:

i Find any stationary points and determine their nature.

ii Determine the points of inflection.

iii Sketch the graph showing stationary points and points of inflection.

iv Write the domain over which the curve is concave up and concave down.

a $f(x) = 3x^2 - x^3 - 4$

b $f(x) = x^3 - 3x^2 - 9x + 11$

c $f(x) = x^3(x - 4)$

d $f(x) = x^2(x^2 - 6x + 12)$

© ClipSAT: By Mr. Mohamed Abdallah +977597688647

13 Consider the function $y = 2(x+5)^3 + 3$.

a Find the y-intercept.

b Find the x-intercept(s).

c Find the stationary points.

d Determine the nature of the stationary points.

e Find any points of inflection and classify them.

f Describe the behaviour of the function as $x \to \pm\infty$.

g Sketch the function on a number plane.

14 Consider the function $y = 2 + 9x - 3x^2 - x^3$.

a Find the y-intercept.

b Find the stationary points.

c Determine the nature of the stationary points.

d Find any points of inflection and classify them.

e Describe the behaviour of the function as $x \to \pm\infty$.

f Sketch the function on a number plane.

15 Consider the function $y = -8(x-1)^2(x+2)$.

a Find the y-intercept.

b Find the x-intercept(s).

c Find the stationary points.

d Determine the nature of the stationary points.

e Find any points of inflection and classify them.

f Describe the behaviour of the function as $x \to \pm\infty$.

g Sketch the function on a number plane.

16 Consider the function $y = (2x+1)(x+2)^4$.

a Find the y-intercept.

b Find the x-intercept(s).

c Find the stationary points.

d Determine the nature of the stationary points.

e Describe the behaviour of the function as $x \to \pm\infty$.

f Sketch the function on a number plane.

© ClipSAT: By Mr. Mohamed Abdallah +977597688647

17 Consider the function $y = \dfrac{8}{x^2+4}$.

a Find the y-intercept.

b Find the x-intercept(s).

c Determine whether the function is odd or even.

d Find the stationary points.

e Determine the nature of the stationary points.

f Describe the behaviour of the function as $x \to \pm\infty$.

g Sketch the function on a number plane.

© ClipSAT: By Mr. Mohamed Abdallah +977597688647